Kanav Atre

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Software engineer and researcher building agentic, LLM-backed data systems. Experience in MCP tooling, semantic query optimization, and workflow automation, with a focus on reliability and performance.

Education

Purdue University West Lafayette, IN
Bachelor of Science Double Major in Data Science & Applied Statistics *Expected Dec 2026*

- **Coursework**: Data Mining & Machine Learning, Intro to AI, Analysis of Algorithms, Relational Databases
- **Involvement**: 2 x presenter at **Purdue's Undergraduate Research Conference**; Purdue LaunchPad **Project Lead**; Student Ambassador for Purdue VIP

Experience

Research Assistant (Under Prof. Chunwei Liu) August 2026 – Now
Purdue University *West Lafayette, IN*

- Studying **cardinality estimation for semantic operators** over unstructured data
- Developing a semantic CE layer for **Palimpzest** with offline catalog and lightweight judge model

Software Engineer Intern May 2025 – Aug 2025
KetteQ *Atlanta, GA*

- Built **agentic retrieval** pipeline serving semantic code search to LLM agents via an **MCP** server
- Designed **evaluation harness** tracking key accuracy measures and agent task success to tune retrieval
- Built a real-time migration pipeline for **10M+ CRM records** from Neon to ClickHouse
- Wrote & Optimized large-scale **PostgreSQL** queries, enabling **sub-second** refresh

Undergraduate ML Researcher Aug 2023 – May 2024
Purdue VIP (Vertically Integrated Projects) *West Lafayette, IN*

- Built a multi-stage object detection pipeline with Faster R-CNN & Multi-task loss, trained on **50K images**
- Implemented core detection components including RPNs, RoI pooling, and NMS for scalable multi-object inference
- Statistically concluded a **120% increase** in post-COVID pedestrian activity and delivered a deployable model

ML Engineer Intern Aug 2022 – Jan 2023
Renzoe Box *Austin, TX*

- Improved **skin-tone detection** algorithms, contributing to the Renzoe-Match engine
- Integrated **YOLOv4** and **dlib-based** facial detection pipelines for **120%** faster image processing
- Wrote web-scraping scripts with **BeautifulSoup**, improving product recommendation accuracy
- Constructed and preprocessed a **1000+ image** dataset for model training

Projects

CodeMap - Semantic Code Search | *MCP, Docker, Kubernetes, PostgreSQL, AWS EC2* Aug 2026

- Developing an MCP-based semantic code search system that uses chunking, vector embeddings, and PostgreSQL/pgvector to enable natural-language code search and repository chat through Claude/Claude Code
- Expanding the system with Kubernetes orchestration, automated retrieval-accuracy evaluation, and multi-repository indexing to improve scalability, search quality, and support for larger codebases.

Relational DBMS From Scratch | *Java, Buffer Manager, Heap File, Clock Replacement* Jan 2026 – Mar 2026

- Built a custom **Buffer Manager** with pin/unpin, dirty tracking, and **Clock replacement** over a fixed frame pool
- Designed a chained hash table mapping page IDs to frame indices for **O(1)** page lookup during pin operations
- Implemented **HeapFile** & **HeapScan** on top of the custom BufMgr and Disk Manager, supporting insert, delete, update & sequential scan
- Resolved circular init between disk & buffer layers via a static registry, and reduced disk writes with **clean-first**

Credify.ai | *Fetch.ai uAgents, GPT-4o, RAG, MV3, Claude* Aug 2025 – Oct 2025

- Built a real-time credibility scoring system for Reddit posts using live context extraction
- Designed a **multi-agent backend** using Fetch.ai uAgents with gateway orchestration and NYT-based retrieval
- Implemented **LLM-based credibility scoring** with JSON schemas and exposed results via REST API
- Built an **MV3 extension** to inject a **React** dashboard using MutationObserver, tab listeners, and cached analysis

Technical Skills

Languages: Python, SQL, Java, R, Bash, C/C++, JavaScript, TypeScript, HTML/CSS
Frameworks: REST APIs, FastAPI, Flask, Node.js, Express, TensorFlow, PyTorch, React, WebSockets, MCP
Tools: Azure, Databricks, AWS (EC2, Lambda), Git, GitHub Actions, CI/CD, Docker, Kafka, Linux
Libraries: Pandas, NumPy, Matplotlib, LangChain, OpenCV, YOLO, pgvector